LUIGI GARCIA

SENIOR GAME PROGRAMMER

luigigamedev@outlook.com • linkedin.com/in/luigigamedev • luigigamedev.github.io

Game programmer with 10+ years of experience shipping games across PC, VR, mobile, and web. Specialized in gameplay systems, architecture, multiplayer netcode, and optimization — primarily in Unreal and Unity.

SKILLS

- Languages: C++ (expert), C# (expert), Python, Java, JavaScript, Lua.
- Engines: Unity (expert), Unreal (expert).
- Software architecture: design patterns, ECS, data-oriented, object-oriented, unit testing.
- Netcode: synchronization, lag compensation techniques, client/server, client prediction.
- Strong math and physics: vectors, matrices, quaternions, dynamic bodies, collisions.
- Optimization and low-level programming: memory, caching, CPU, GPU.
- Graphics programming: OpenGL, shader programming.
- Game AI: state machines, behavior trees, pathfinding, steering behaviors.

EXPERIENCE

Independent Game Developer [2025 - Present]

- Developed and published Car Controller: Arcade & Simcade on the Unity Asset Store.
- Developing a custom C++/OpenGL game engine from scratch as a learning journey into low-level rendering and engine systems.
- Developing a multiplayer game in Unreal.

Lead Game Programmer at Hololabs, Victoria, Canada [2023 - 2024]

- Led a team of 3 programmers; managed planning, code reviews, and cross-team alignment with design, art, and production.
- Advocated for scope reduction early, negotiating milestones that delivered a successful Phantasms VR closed alpha submission (CMF).
- Implemented VR locomotion, interactables, grab/throw, and equipment holstering; integrated multiplayer replication for all gameplay systems.
- Wrote Unreal networking and gameplay framework documentation; mentored programmers through recorded pair programming sessions.

Senior Game Programmer at Ludia, Montreal, Canada [2019 - 2022]

- Improved UI rendering and runtime memory performance by ~50% across Disney and DC mobile projects; wrote technical documentation on rendering, batching, and memory optimization adopted company-wide.
- Implemented park builder mechanics, including camera, structure placement, and zoomed-in object mode.

Lead Game Programmer at People Corp Gaming, Montreal, Canada [2017 - 2018]

- Led the development team; coordinated with design, art, and QA across departments; consulted directly with the CEO on technical direction.
- Implemented the match3 engine for Fuzzy Critters PvP Match3.
- Built a Lua scripting game library on top of Unity, enabling weekly minigame delivery via asset bundles, without app updates.

Senior Game Programmer at Pandora Game Studio, Rio de Janeiro, Brazil [2015 - 2016]

- Implemented player movement and combat for Bushido Saga: Nightmare of the Samurai.
- Built a component-based items system enabling designers to create and mass-edit items, consumables, and equipment from Google Sheets.
- Built a Unity editor tool to bake NavMesh from collider geometry instead of mesh renderers, improving navmesh accuracy on non-standard surfaces.

Game Programmer at Izyplay Game Studio, Pelotas, Brazil [2013 - 2015]

- Implemented combat mechanics, stages, and boss fights for Apocalypse: Party's Over.
- Shipped multiple flash web games.

TOP PROJECTS

Car Controller: Arcade & Simcade - Asset Store Page
A custom car controller for arcade-style games. Published on the Unity Asset Store.

Phantasms VR - Hololabs - Page
A coop multiplayer VR game featuring ghost-capturing combat, exploration, and puzzles. Closed alpha submitted to CMF (Canada Media Fund). Built in Unreal with C++ and VRExpansion Plugin.

Bushido Saga: Nightmare of the Samurai - Pandora Game Studio - Steam Page
An action-adventure RPG game featuring a dynamic combat system with a versatile arsenal of melee and ranged weapons. Released on Steam, Google Play, and Apple. Built in Unity with C#.

Apocalypse: Party's Over - Izyplay Game Studio - Steam Page
A 2D side-scrolling beat'em up game featuring the cartoons of Mundo Canibal. Released on Steam. Built in Unity with C#.

EDUCATION

Major in Software Engineering, FATEC SENAC Pelotas [2014]

- Scholarship — R&D lab in medical image processing for bone X-ray interpretation.